Rajarata University of Sri Lanka

Department of Computing

COM3401 – Data Communication and Networking
Laboratory 01 – Introduction Cisco Packet Tracer and Designing & configuring LANs
Time: 2 Hours

Intended Learning Outcomes (ILOs)

By the end of this practical, students should be able to:

1. Getting familiar with Cisco Packet Tracer interface and Designing Basic topologies
2. Designing and configuring a LAN

Pre-Practical Preparation

1. **Reading/Resources**: Refer to the topics covered in Lecture 1 and Lecture 2

Practical Objective:

Explain the interface and devices available in Cisco Packet Tracer. Make students aware of the devices and connection methods available. Allow students to design and configure LANs based on scenarios given. Students will learn how to design and configure basic networks according to the list of devices given to them.

Practical Activity

Activity 1:

Exploring the Cisco Packet Tracer application

- Objectives: Making students aware of the functionalities provided in Packet Tracer.
- Instructions:
    1. Download and install Packet Tracer
    2. Launch Packet Tracer
    3. Explore the device categories available
    4. Drag and drop a device and explore the available device options
    5. Explore Logical and Physical work area
    6. Explore Realtime mode and Simulation mode
    7. Design a star topology using following devices:

        - 1 x Switch (2960)
        - 5 x PC
        - Copper straight-through cables

Activity 2:

Connecting two computes using a switch

- Objectives: Designing and configuring a basic LAN
- Instructions:
    1. Create a new project
    2. Add following devices:
        - 1 x switch (2960)
        - 2 x PC
    3. Connect two PCs to the switch using Copper straight-through cables
    4. Configure PC0 with following IP configuration
        - IP: 192.168.1.10
        - Mask: 255.255.255.0
    5. Configure PC1 with following IP configuration
        - IP: 192.168.1.20
        - Mask: 255.255.255.0
    6. Verify the connectivity
        - Ping PC1 from PC0
        - Ping PC1 from PC0

Activity 3:

Designing a classroom network

Objective: Designing and configuring a LAN for a real-world scenario

- Instructions:
    1. Create a new project
    2. Add following devices:
        - 1 x switch (2960)
        - 5 x PC
        - 1 x server

3. Connect PCs and server to the switch using Copper straight-through cables
4. Configure PC0 with following IP configuration
    - IP: 192.168.1.10
    - Mask: 255.255.255.0
5. Configure PC1 with following IP configuration
    - IP: 192.168.1.20
    - Mask: 255.255.255.0
6. Configure PC2 with following IP configuration
    - IP: 192.168.1.30
    - Mask: 255.255.255.0
7. Configure PC3 with following IP configuration
    - IP: 192.168.1.40
    - Mask: 255.255.255.0
8. Configure PC4 with following IP configuration
    - IP: 192.168.1.50
    - Mask: 255.255.255.0
9. Configure server with following IP configuration
    - IP: 192.168.1.60
    - Mask: 255.255.255.0
10. Verify the connectivity
    - Ping the server from PC0, PC1, PC2, PC3 and PC4

Activity 4:

Designing a complex LAN

Objective: Designing and configuring a complex LAN for a real-world scenario

- Instructions:
    1. Create a new project
    2. Add following devices:
        - 2 x switch (2960)
        - 5 x PC
        - 1 x server
    3. Connect PC0, PC1 And server to switch0 using Copper straight-through cables
    4. Connect PC2, PC3 And PC4 to switch1using Copper straight-through cables

5. Configure the network using the knowledge you have gained so far
6. Verify the connectivity
    - Ping the server from PC0, PC1, PC2, PC3 and PC4